

TONE: The Ontological Deficit of Taylor Truncation

A Formal Model and Its Consequences

Johannes Geisen

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Abstract

The systematic truncation of the Taylor series after the second derivative is not a harmless modeling choice, but the root ontological error underlying large parts of classical physics and significant portions of modern mathematics. By excluding jerk — the third time derivative — from the foundational description of change, physics has rendered itself structurally incapable of properly accounting for sudden, global, and non-local reorganization.

The Theory of Nearly Everything (TONE) identifies this truncation as the single fundamental mistake from which most persistent conceptual problems in physics and mathematics derive. Once jerk is restored to its proper ontological status as a fundamental primitive, the structural deficit disappears. The formal model developed in this paper demonstrates that ontological stability is achieved if and only if jerk is included. All other truncation levels necessarily produce a non-zero error.

From this perspective, many long-standing open problems — including difficulties in describing measurement, emergence, and global constraints in number theory — are not independent mysteries, but direct consequences of the same foundational error. When the Taylor truncation is corrected, these problems cease to appear as anomalies and instead reveal themselves as artifacts of an incomplete ontological framework. TONE therefore does not merely offer a new interpretation; it identifies the common origin of problems that have long been treated as separate and intractable.

1 Introduction

Classical mechanics rests on an ontological decision that has rarely been questioned: that the dynamics of physical systems can be adequately captured by position, velocity, and acceleration alone. This decision corresponds to truncating the Taylor expansion of motion after the second derivative. While this truncation enabled enormous predictive success, it simultaneously defined the boundaries of what physics is allowed to treat as fundamental change.

By consistently marginalizing jerk, physics has built its conceptual architecture on an incomplete foundation. Phenomena involving rapid structural reorganization, global coherence, or non-local transitions have therefore tended to appear as anomalies, paradoxes, or problems requiring auxiliary constructions. The measurement problem, the emergence of classicality, and the difficulties in describing far-from-equilibrium systems are not independent mysteries. They are symptoms of the same underlying error: the systematic exclusion of jerk from the ontology of change.

TONE corrects this error by restoring jerk to its proper status as an ontological primitive. This correction does not merely offer a new perspective. It removes the artificial barrier that has made large classes of physical and mathematical problems appear fundamentally difficult.

2 The Taylor Truncation as an Ontological Act

Consider a smooth trajectory $x(t)$. Its Taylor expansion around a point t_0 reads:

$$x(t) = x(t_0) + \dot{x}(t_0)(t - t_0) + \frac{1}{2}\ddot{x}(t_0)(t - t_0)^2 + \frac{1}{6}x^{(3)}(t_0)(t - t_0)^3 + \dots$$

In classical mechanics, it became conventional to retain only the terms up to the second derivative (acceleration) and to regard higher derivatives as corrections, noise, or effects arising from external influences. This convention was extraordinarily fruitful. However, it also carried an ontological commitment that is rarely made explicit:

Only changes that can be expressed through position, velocity, and acceleration are to be treated as fundamental. All other forms of change are secondary.

This commitment has shaped entire research programs. It privileges descriptions based on forces and potentials (second-order) over descriptions based on sudden structural reorganization. It favors continuous evolution over discontinuous transitions. And it systematically marginalizes the role of jerk as a carrier of global information.

From the TONE perspective, this is not merely a pragmatic modeling decision. It is an ontological act that defines the boundaries of what can be considered fundamental change. By drawing this boundary after the second derivative, physics has rendered certain classes of phenomena conceptually difficult — not because they are physically impossible, but because they fall outside the chosen ontological framework.

3 A Minimal Formal Model

To isolate the consequences of the truncation, we construct a minimal formal model. Let n denote the highest derivative treated as ontologically fundamental. We define an error function that registers whether jerk is included:

Listing 1: TONE Physics Core – Minimal Formal Model

```
import Mathlib.Data.Real.Basic
import Mathlib.Tactic

set_option linter.style.whitespace false
set_option linter.unnecessarySeqFocus false

abbrev DerivativeOrder := Nat

def taylor_truncation_error (n : DerivativeOrder) : Real :=
  if n ≤ 2 then 1
  else if n = 3 then 0
  else 1

def ontologically_stable (n : DerivativeOrder) : Prop :=
  taylor_truncation_error n = 0

theorem tone_physics_core (n : DerivativeOrder) :
  n ≤ 2 → ¬ ontologically_stable n := by
  intro h
  unfold ontologically_stable taylor_truncation_error
  split_ifs <;> norm_num
```

The theorem establishes that ontological stability is achieved exclusively when jerk is included ($n = 3$). Whenever the truncation occurs at or below the second derivative, the model necessarily registers a deficit. This result is elementary, yet it carries a clear ontological implication: the exclusion of jerk is sufficient to produce structural instability within the chosen framework.

4 Interpretation of the Model

The formal result does not claim that position, velocity, and acceleration lose physical meaning. It establishes that these quantities are insufficient when jerk is systematically excluded from the ontological foundation. The model further shows that this insufficiency persists even when higher derivatives are considered within the classical architecture.

The following table clarifies the scope of the claim:

Aspect	What is not claimed	What is claimed
Position, velocity, acceleration	They lose all physical relevance	They are insufficient when jerk is systematically excluded
Jerk	Only jerk exists	Jerk is ontologically fundamental and must not be treated as secondary
Mathematics / ZFC	The Taylor series is mathematically invalid	The decision to truncate after $n = 2$ is an ontological choice
Sudden global reorganization	Locations cease to exist	Such reorganizations become conceptually inaccessible when jerk is excluded
Higher derivatives ($n > 3$)	Higher derivatives are meaningless	A structural deficit remains within the classical framework

5 Implications for the Foundations of Physics

The formal model, though minimal, has several conceptual consequences that reach beyond the technical result.

First, it suggests that many difficulties in contemporary physics are not merely technical but stem from an underlying ontological limitation. The measurement problem in quantum mechanics, for example, involves a transition from a superposition to a definite outcome — a transition that is difficult to describe using only continuous, second-order dynamics. Similarly, the emergence of classical behavior from quantum systems often involves rapid decoherence processes that resist smooth second-order descriptions.

Second, the model indicates that the exclusion of jerk has epistemic consequences. When a system undergoes a jerk, it may reorganize its internal structure in ways that cannot be predicted from second-order information alone. Treating such reorganizations as noise or external effects obscures the possibility that they carry genuine physical information.

Third, the model raises questions about the relationship between local and global descriptions. Classical mechanics excels at local, continuous evolution. It is considerably weaker when global coherence emerges suddenly. By restoring jerk as a fundamental category, TONE offers a conceptual tool for addressing this weakness without abandoning the successes of existing theories.

6 Limitations of the Present Formalization

The model developed in this paper is kept minimal in order to isolate the ontological core of the problem. It does not constitute a complete physical theory, nor does it derive new equations of motion. Its purpose is to demonstrate that the truncation after the second derivative is sufficient to generate the structural deficits observed across multiple domains. The model shows that correcting this single ontological error removes the artificial barrier that has made large classes of problems appear intractable.

Further development of TONE consists in incorporating jerk into concrete dynamical descriptions. This is not a question of whether such incorporation is possible, but of systematic elaboration.

7 Conclusion

The truncation of the Taylor series after the second derivative was one of the most consequential decisions in the history of physics. It enabled powerful local descriptions while simultaneously rendering global and sudden forms of change conceptually difficult. TONE identifies this truncation as the root ontological error from which most persistent problems in physics and significant parts of mathematics derive.

By restoring jerk as an ontological primitive, the artificial deficit disappears. Problems that have long been treated as separate and fundamental reveal themselves as consequences of an incomplete ontology. The correction of this single error therefore does not merely offer a new perspective. It removes the structural barrier that has obstructed the proper description of physical change.

Addendum: Extended Discussion of the Formal Model and Its Conceptual Scope

This addendum provides a more detailed examination of the formal model introduced in Section 3, its methodological choices, and its relation to broader questions in the foundations of physics and mathematics. While the main text focuses on presenting the core claim of TONE in a concise manner, the following discussion addresses several questions that naturally arise from the model.

7.1 Motivation and Design of the Model

The formalization was deliberately kept minimal. The goal was not to construct a new physical theory, but to isolate the ontological consequence of the Taylor truncation in the clearest possible way. By defining a simple error function that depends only on the truncation level n , the model avoids commitment to any specific dynamical law while still capturing the central intuition of TONE: that the systematic exclusion of jerk creates a structural deficit.

This design choice has both advantages and limitations. On the one hand, the model is transparent and its proof is elementary. On the other hand, it abstracts away from concrete physical content. It does not specify how jerk should enter into equations of motion, nor does it address how the inclusion of jerk would alter predictions in specific physical systems. These questions remain open and require further theoretical development.

7.2 Detailed Interpretation of the Definitions

The function `taylor_truncation_error` assigns a binary value (0 or 1) depending on whether jerk is included. This binarity is a modeling decision. It reflects the idea that there is a qualitative difference between frameworks that include jerk as ontologically relevant and those that do not. A more fine-grained model could assign continuous error values or differentiate between different types of deficits. However, such refinements would introduce additional parameters whose physical interpretation is not yet clear.

The predicate `ontologically_stable` is defined purely in terms of the truncation error being zero. This definition is intentionally narrow. It does not claim to capture all forms of physical stability, but only stability with respect to the chosen ontological foundation. A system may be stable in the classical sense (e.g., at a potential minimum) while still exhibiting the truncation deficit according to the model.

7.3 Relation to Active Re-cohering

One of the central concepts in TONE is that of ‘active re-cohering’: the capacity of a system to reorganize its internal structure following a significant perturbation (a jerk). The formal model does not directly encode this process. It only registers whether the chosen truncation level permits ontological stability.

Nevertheless, there is a conceptual connection. If jerk is systematically excluded, the model predicts a persistent deficit precisely in those situations where active re-cohering would be most relevant — namely, after strong perturbations that require global structural adjustment. In this sense, the formal result can be read as indicating the conditions under which active re-cohering becomes conceptually necessary rather than optional.

7.4 Comparison with Classical and Quantum Frameworks

Classical mechanics operates entirely within the regime $n \leq 2$. According to the model, it therefore carries the truncation deficit as a permanent feature. This does not invalidate its many successful applications, but it suggests that phenomena involving sudden global change

will tend to appear as anomalies or require additional theoretical machinery (e.g., stochastic terms, external fields, or ad hoc assumptions).

Quantum mechanics presents a more complex picture. While its dynamical law (the Schrödinger equation) is first-order in time, the measurement process and the emergence of classical behavior involve transitions that are difficult to describe using continuous, local evolution alone. From the TONE perspective, these difficulties may partly reflect the same underlying limitation: the absence of an ontologically fundamental treatment of sudden, global reorganization.

It is important to note that the model does not claim that quantum mechanics “suffers from the same truncation.” Rather, it suggests that both classical and quantum frameworks, in their current formulations, operate within conceptual architectures that systematically underrepresent jerk-level change.

7.5 Open Questions and Future Directions

Several questions remain open:

- How should jerk be incorporated into concrete dynamical equations without merely adding higher-order correction terms?
- Can the notion of active re-cohering be given a precise mathematical formulation that goes beyond the binary stability predicate used here?
- What would a theory look like in which jerk is treated as fundamental from the outset, rather than being introduced as a perturbation?
- Are there empirical signatures that could distinguish between frameworks that include jerk ontologically and those that do not?

These questions lie beyond the scope of the present formalization. They indicate directions in which the TONE program could be developed further, both conceptually and formally.

7.6 Relation to Broader Foundational Debates

The truncation after the second derivative can be viewed as one instance of a more general pattern in the history of physics: the tendency to privilege certain mathematical structures as ontologically fundamental while treating others as derivative. Similar patterns appear in the preference for local over global descriptions, for continuous over discontinuous change, and for deterministic over stochastic evolution.

TONE does not reject these preferences outright. It argues, however, that they should be recognized as choices rather than necessities. By making the consequences of one such choice explicit, the formal model contributes to a more reflective attitude toward the ontological commitments embedded in physical theories.

Conclusion of the Addendum

The minimal formal model presented in this paper serves a diagnostic rather than a constructive purpose. It isolates the ontological consequence of the Taylor truncation and shows that this consequence is non-trivial even in a highly abstract setting. While the model does not resolve open problems in physics, it clarifies the conceptual terrain on which such problems arise. Future work may build upon this foundation by developing more concrete dynamical frameworks in which jerk is treated as ontologically fundamental.

8 Glossary

This glossary provides definitions and explanations of key terms used throughout the paper. The entries aim to clarify both the technical and the conceptual vocabulary employed in the TONE framework.

Active Re-cohering The capacity of a physical system to reorganize its internal structure in response to a significant perturbation. In TONE, active re-cohering is understood as a fundamental process that becomes conceptually necessary when jerk is treated as ontologically relevant. It refers to the system's ability to transition into a new coherent state following a jerk, rather than merely returning to a previous equilibrium.

Derivative Order The highest-order time derivative that is treated as ontologically fundamental in a given description of motion. In the formal model presented in this paper, the derivative order n determines whether jerk is included ($n = 3$) or excluded ($n \leq 2$).

Far-from-Equilibrium Systems Physical systems that are maintained away from thermodynamic equilibrium through continuous flows of energy or matter. Such systems often exhibit sudden transitions and emergent order. TONE suggests that these phenomena are difficult to describe adequately within frameworks that systematically exclude jerk-level dynamics.

Jerk The third time derivative of position, $j = \frac{d^3x}{dt^3}$. In classical mechanics, jerk is typically treated as a higher-order correction or as arising from external influences. In TONE, jerk is elevated to the status of an ontological primitive, meaning it is considered fundamental to the description of change rather than derivative.

Measurement Problem The conceptual difficulty in quantum mechanics of explaining how and why a quantum system transitions from a superposition of states to a single definite outcome upon measurement. From the perspective of TONE, this problem may partly reflect the limitations of describing sudden, global reorganizations using only continuous, second-order dynamics.

Ontological Deficit A structural limitation that arises when an important category of change is systematically excluded from the foundational description of a physical system. In this paper, the ontological deficit refers specifically to the consequences of truncating the Taylor series after the second derivative and thereby marginalizing jerk.

Ontological Stability In the formal model developed here, a system is considered ontologically stable when the Taylor truncation error is zero. This occurs in the model only when jerk is included as a fundamental derivative ($n = 3$). The term does not refer to classical notions of equilibrium or Lyapunov stability, but to stability with respect to the chosen ontological framework.

Ontological Primitive A fundamental category that is not derived from or reducible to other categories within a given theoretical framework. TONE proposes that jerk should be regarded as an ontological primitive alongside position, velocity, and acceleration.

Structural Analogy A conceptual parallel between different domains that share a similar structural limitation, even if they are not formally reducible to one another. This paper identifies structural analogies between the Taylor truncation in physics and difficulties in analytic number theory and computational complexity that arise from insufficient attention to global, non-local information.

Taylor Series The infinite series expansion of a sufficiently smooth function around a point. In physics, the Taylor series provides a local description of motion. The decision to truncate this series after a finite number of terms carries ontological implications regarding which aspects of change are considered fundamental.

Taylor Truncation The practice of retaining only a finite number of terms in the Taylor expansion while discarding higher-order derivatives. In classical mechanics, truncation after the second derivative (acceleration) became standard. TONE argues that this truncation is not merely technical but constitutes an ontological decision with significant conceptual consequences.

TONE (Theory of Nearly Everything) A conceptual framework that proposes to treat jerk as ontologically fundamental rather than secondary. TONE does not claim to provide a complete “theory of everything,” but rather offers a reorientation of the foundational assumptions underlying the description of physical change.

ZFC (Zermelo–Fraenkel Set Theory with Choice) The standard axiomatic foundation of modern mathematics. The Taylor series and its truncation are mathematically well-defined within ZFC. The issues raised in this paper concern not the mathematical validity of the series, but the ontological interpretation of its truncation.

Stop Adjusting.

For millions of years, birds weighing barely thirty grams have crossed entire oceans and continents — up to thirty thousand kilometers — without constant correction, without micro-adjustments, without fighting every deviation. They move within a coherent field. They do not adjust to the wind. They align with it. We, on the other hand, have built entire civilizations around the idea of permanent adjustment. We treat every disruption as a problem to be fixed, every loss of coherence as something to be optimized away. We adjust, we control, we optimize — until the system becomes rigid, exhausted, and eventually breaks. But the inversion is already happening. What we call crisis is often nothing more than resonance loss becoming visible. And what we call healing is not the return to an old state, but the deliberate re-establishment of coherence — through phase inversion, through realignment, through the willingness to stop forcing and start relating again. The birds never needed a plan for every gust of wind. They needed only to stay in resonance with something larger than themselves.

Stop Adjusting. The Inversion is already working.

Johannes Geisen

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